ST ENGINEERING

HOW TO TRAIN
CNN

LEARN NOW →

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DATA
PREPARATION

```
dataset/
├── train/
│   ├── cats/
│   │   ├── cat_001.jpg
│   │   ├── cat_002.jpg
│   │   └── ...
│   └── dogs/
│       ├── dog_001.jpg
│       ├── dog_002.jpg
│       └── ...
└── test/
    ├── cats/
    │   ├── cat_test_001.jpg
    │   └── ...
    └── dogs/
        ├── dog_test_001.jpg
        └── ...
```

80% OF IMAGES IS TRAIN DATA

20% OF IMAGES IS TEST DATA

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PREPARE INPUT

ANALYZE FOLDER OR CSV FILE → MAPPING COLUMN WITH FOLDER STRUCTURE → PROCESS INTO IMAGES AND LABELS

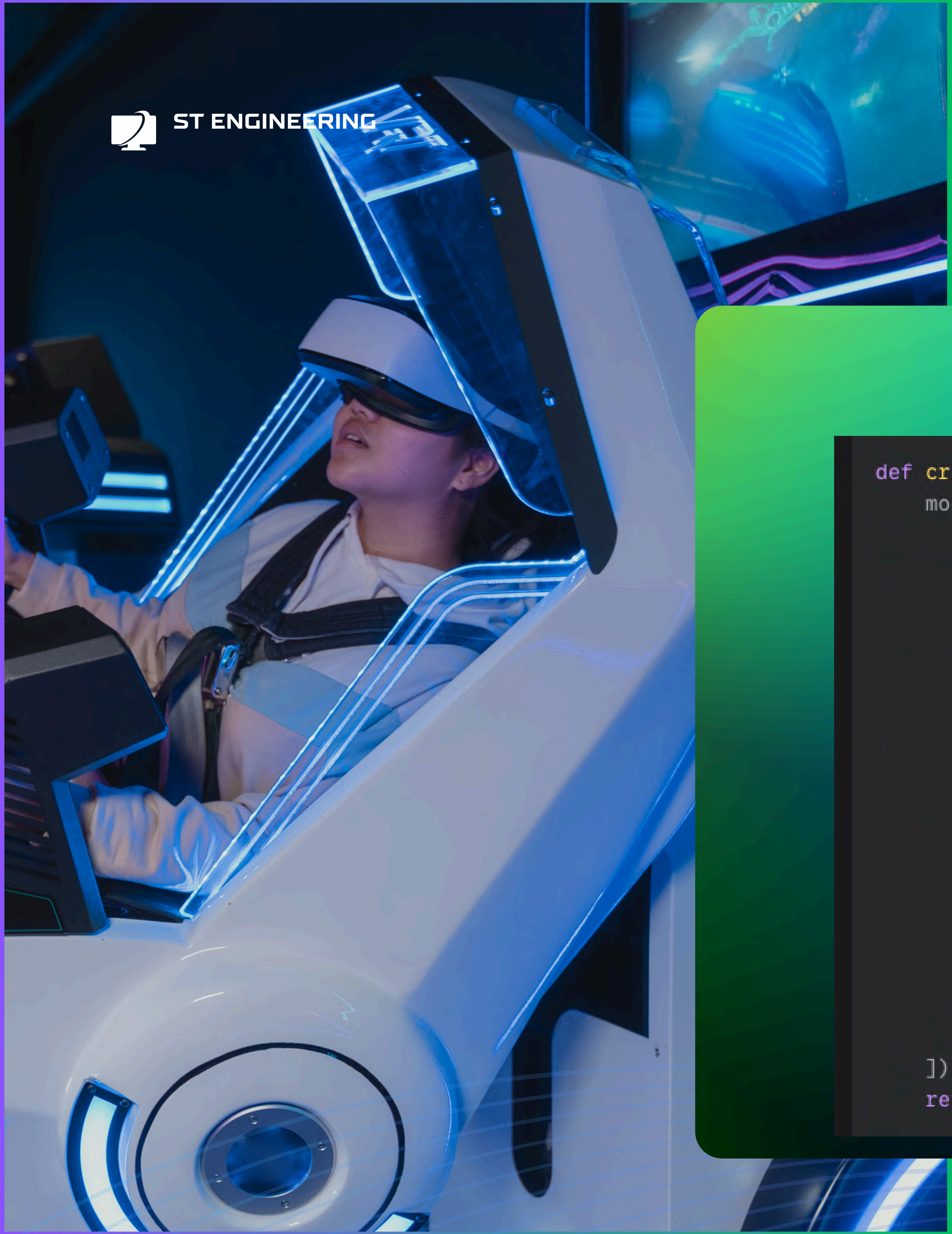
PREPOCESS IMAGE → MAPPING COLUMN WITH FOLDER STRUCTURE

this step will be used for prediction phase

PROCESS INTO IMAGES AND LABELS →

processed Image 1
processed Image 2
processed Image 3
processed Image 4

label 1
label 2
label 3
label 4



Define CNN

```python
def create_sequential_cnn(input_shape, num_classes):
    model = keras.Sequential([
        # First Convolutional Block
        layers.Conv2D(32, (3, 3), activation='relu', input_shape=input_shape),
        layers.MaxPooling2D((2, 2)),

        # Second Convolutional Block
        layers.Conv2D(64, (3, 3), activation='relu'),
        layers.MaxPooling2D((2, 2)),

        # Third Convolutional Block
        layers.Conv2D(128, (3, 3), activation='relu'),
        layers.MaxPooling2D((2, 2)),

        # Flatten and Dense Layers for Classification
        layers.Flatten(),
        layers.Dense(128, activation='relu'),
        layers.Dropout(0.5), # Add dropout for regularization
        layers.Dense(num_classes, activation='softmax') # Output layer for classifi
    ])
    return model
```

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Training Process

CONVERT IMAGES AND LABEL INTO X_TRAIN AND Y_TRAIN

CONVERT IMAGES AND LABEL INTO X_TRAIN AND Y_TRAIN

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BATCH_SIZE

HIGHER

- COMPUTATIONAL EFFICIENCY AND FASTER TRAINING PER EPOCH
- MORE STABLE GRADIENT ESTIMATES
- BETTER USE OF MEMORY (IN SOME CASES)

LOWER

When to Consider a Lower BATCH_SIZE:

- When generalization performance is a top priority.
- When memory is limited.
- When you want the regularizing effect of noisy gradients to help avoid overfitting and escape local minima.

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PREDICT **IMAGE**

Preprocess

This step is the same with preparation data

Predict

- Predict
- Get the position max value of X axis
- The result is the label at that position

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ASSIGNMENT

+123-456-7890

WWW.REALLYGREATSITE.COM

HELLO@REALLYGREATSITE.COM

123 ANYWHERE ST., ANY CITY, ST 12345

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THANK YOU